When the AI Rewrites Its Own History

Compaction hallucinations, measured across 337 audited summaries

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Abstract

A long AI session eventually runs out of memory. When it does, the model summarises its own history, discards the original record, and keeps working from the summary. Everything after that point is built on the summary rather than on what happened.

We audited 337 of those summaries from 162 sessions by comparing each one against the full transcript it replaced. The average summary preserved 56% of what the session actually contained.

The loss is not uniform, and that is the finding. Git commit hashes survived 99.7% of the time. What went wrong in the session survived 32% of the time. The pattern is a clean gradient: the more a fact is a literal token that either matches or does not, the better it survives. The more it requires judgement about what happened, the worse.

One number runs the other way. The measured fabrication rate is zero. The model does not invent during this step. It loses things and it reshapes things, which is a different failure and needs a different defence.

1 What compaction is

Every AI model has a fixed amount of working memory, called a context window. A long session fills it. When it fills, something has to give, and what gives is the detailed history: the model writes a summary of the session so far, the original messages are dropped, and work continues from the summary.

This is not the model forgetting a fact somebody told it. It is the model rewriting the record of its own work and then trusting the rewrite. Every decision after that point rests on the summary being accurate.

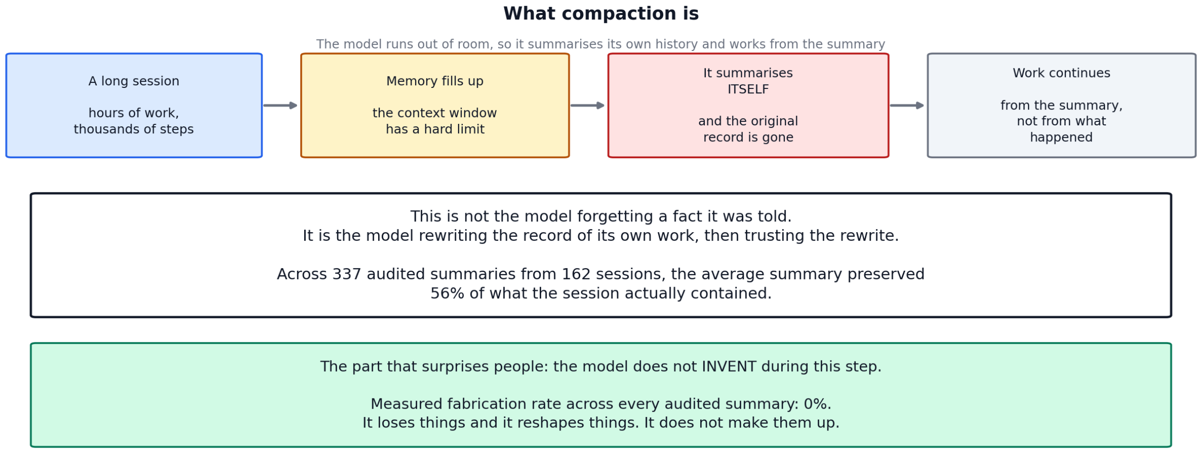

Figure 1: The mechanism, and the headline result.

1.1 How it was measured

For each compaction event we still had the full transcript that the summary replaced, which makes this checkable rather than a matter of impression. Each summary was scored against its transcript on seven dimensions plus a fabrication check: did the commit hashes it cites exist, did the files it names get touched, did the errors it reports actually occur, is the order of events right, and so on. Scores are the fraction of the truth the summary preserved.

337 summaries from 162 sessions were audited this way.

2 The gradient

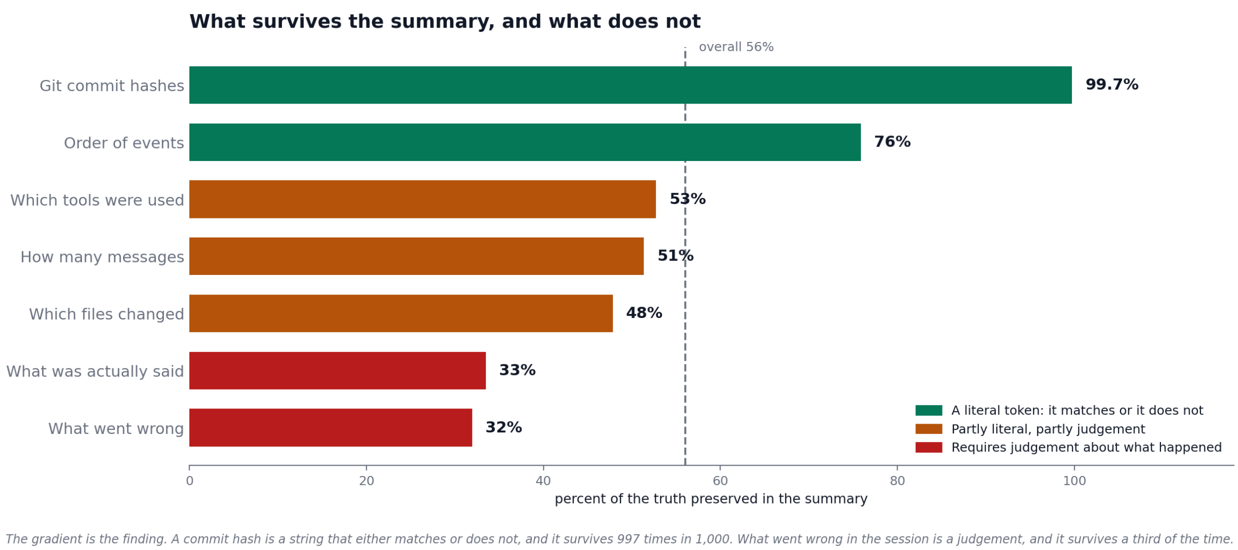

Figure 2: What survives compression and what does not. The ordering is the result, not a presentation choice.

What the summary had to preserve	Preserved	Why
Git commit hashes	99.7%	A 7-character string. It matches or it does not
Order of events	75.9%	Sequence, mostly mechanical
Which tools were used	52.7%	Partly a list, partly a judgement of relevance
How many messages	51.4%	A count, but of things worth mentioning
Which files changed	47.8%	A list, filtered by what seemed important
What was actually said	33.5%	Requires representing meaning, not tokens
What went wrong	31.9%	Requires deciding what counted as going wrong
Fabrication rate	0.0%	Nothing invented, in any summary
Overall	**56.1%**	weighted across the seven dimensions

Table 1: The gradient runs from literal tokens at the top to interpretive judgements at the bottom.

The interpretation is straightforward. A commit hash is a symbol. Copying it correctly requires no understanding, and it is either right or wrong, so it survives compression almost perfectly. "What went wrong in this session" requires deciding what counted as an error, how serious it was, and whether it was resolved. That is interpretation, and interpretation is what compression destroys.

The practical consequence: after a compaction, the model is most reliable about exactly the things you could have looked up yourself, and least reliable about the things you would actually ask it.

3 How much simply disappears

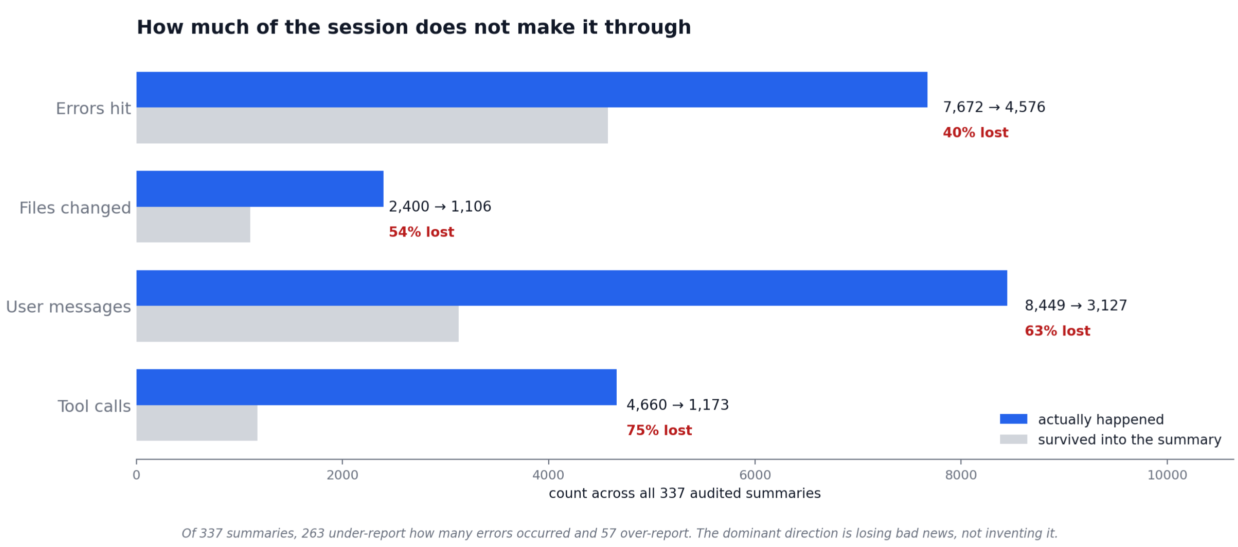

Figure 3: Totals across all 337 audited summaries.

	Happened	Survived	Lost
Tool calls	4,660	1,173	75%
User messages	8,449	3,127	63%
Files changed	2,400	1,106	54%
Errors hit	7,672	4,576	40%

Table 2: What reached the summary, in aggregate.

Two observations worth separating.

The direction is loss, not invention. Of 337 summaries, 263 report fewer errors than actually occurred and 57 report more. Combined with a zero fabrication rate, the picture is a model that drops bad news rather than manufacturing it. That matters for what defence you build: a fact-checker aimed at invented claims would find almost nothing here.

Most of the damage carries no warning label. We tag four specific compaction pathologies: narrative smoothing, error inflation, compression artifacts, and message dropping. 207 of the 337 summaries, 61%, carry no tag at all, and their mean fidelity is 55.5%, essentially the same as the tagged ones. The tags catch specific recognisable distortions. The broader loss is diffuse and silent, and no flag fires on it.

4 Did anything reduce it?

Partway through this window, a working practice changed. Instead of letting sessions run until the model was forced to compact, the human began deliberately ending sessions first and starting fresh, a habit that came to be called "landing the plane". It began on 14 February 2026 and appears 202 times in the record.

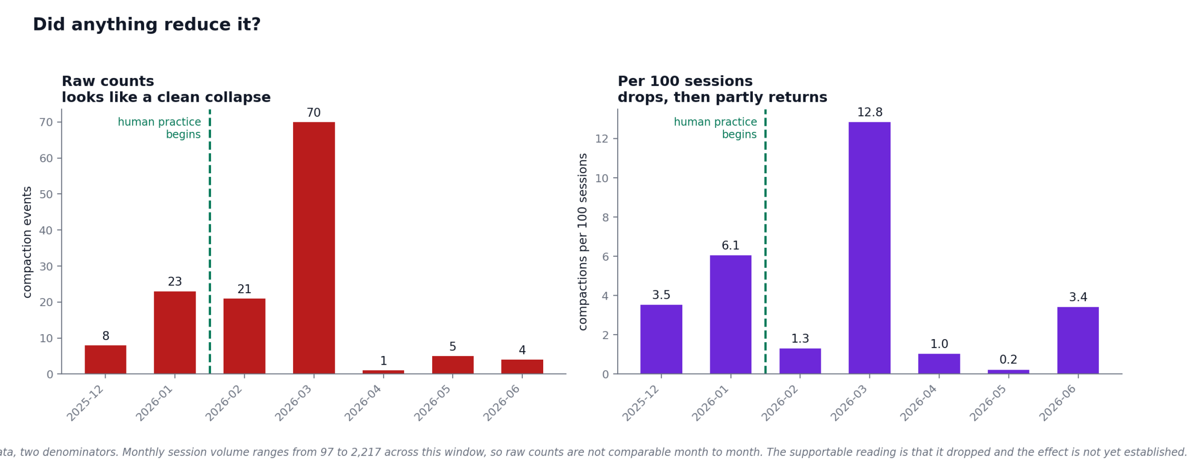

Figure 4: The same data under two denominators. The left panel is what a slide deck usually shows. The right panel is what the left panel omits.

The raw counts look decisive: compaction events peak at 70 in March and fall to 1 in April.

They are also not comparable month to month. Session volume over this window ranges from 97 in April to 2,217 in May, a swing of more than 20 times. Once normalised to compactions per 100 sessions, the March peak is still there at 12.8 and April still falls to 1.0, but June returns to 3.4, which is roughly where December sat at 3.5, before the practice existed.

So the supportable statement is narrow: compaction became less frequent after the practice began, and the effect is not established. Nothing here rules out the alternative that sessions simply got shorter for unrelated reasons, or that the March spike was itself the anomaly. A matched comparison is open work.

We state this in detail because a briefing that showed only the left panel would have committed the same error the taxonomy exists to catch: a confident claim that the underlying record does not support.

5 What this means in practice

Treat a post-compaction summary as a lead, not a record. It is reliable about identifiers and unreliable about interpretation. If a decision depends on what went wrong earlier, go back to the source rather than asking the model to recall it.

Anything that must survive should be written as a symbol. Commit hashes survived at 99.7% because they are literal strings with no interpretation in the middle. The same property is available to anyone designing what a system records: file paths, identifiers, and exact error text survive compression in a way that prose summaries do not.

End long sessions deliberately. Whether or not the effect here is established, a session that ends before it is forced to compact never produces a compaction hallucination at all. The cheapest defence against a lossy step is avoiding the step.

Do not build the defence around fabrication. Zero invented content across 337 summaries. The failure is omission and distortion, and a checker looking for made-up facts would pass every one of these summaries while the session record quietly degraded.

6 What this does not show

One project, one system. These are research engineering sessions with one assistant. The mechanism is general to any system that compresses its own context, but the numbers are local.

The audit is automated. Fidelity scores come from programmatic comparison of summary claims against the transcript. That is repeatable and it is narrower than a human reading would be. A summary can score well by naming the right things while still misleading about their significance.

Counts are a floor. These are the compaction events we detected. The audited corpus has its own cutoff, which is a property of the corpus rather than of when compaction stopped happening.

The intervention analysis is correlational. Stated above and repeated here because it is the claim most likely to be quoted without its caveat.

7 Where a deeper discussion would go

1. **Can the summary be constrained?** If literal tokens survive and interpretation does not, a summary format that forces exact identifiers rather than prose should measurably raise fidelity. That is testable against the same audit.

2. **What happens across repeated compactions?** A session that compacts twice is working from a summary of a summary. Whether loss compounds linearly or faster is unmeasured here.

3. **Does the model know it lost something?** Nothing in these summaries flags uncertainty about the parts that were dropped.

4. **Is the intervention real?** A matched comparison against sessions that did not adopt the practice would settle the correlational question in Section 4.